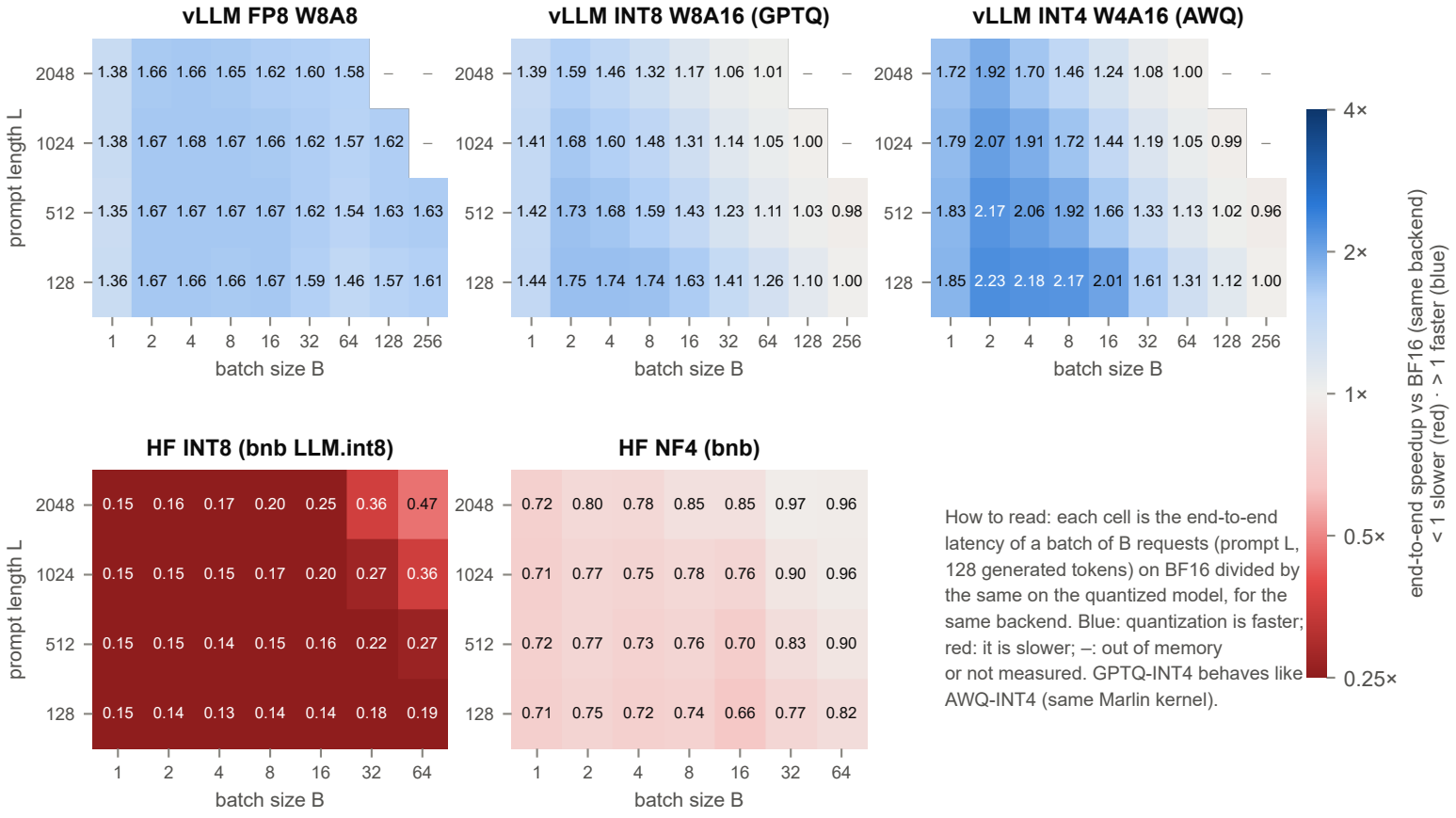

vLLM FP8 W8A8
2048 1.38 1.66 1.66 1.65 1.62 1.60 1.58 – –
1024 1.38 1.67 1.68 1.67 1.66 1.62 1.57 1.62 –
512 1.35 1.67 1.67 1.67 1.67 1.62 1.54 1.63 1.63
128 1.36 1.67 1.66 1.66 1.67 1.59 1.46 1.57 1.61
1 2 4 8 16 32 64 128 256
batch size B
vLLM INT8 W8A16 (GPTQ)
2048 1.39 1.59 1.46 1.32 1.17 1.06 1.01 – –
1024 1.41 1.68 1.60 1.48 1.31 1.14 1.05 1.00 –
512 1.42 1.73 1.68 1.59 1.43 1.23 1.11 1.03 0.98
128 1.44 1.75 1.74 1.74 1.63 1.41 1.26 1.10 1.00
1 2 4 8 16 32 64 128 256
batch size B
vLLM INT4 W4A16 (AWQ)
2048 1.72 1.92 1.70 1.46 1.24 1.08 1.00 – –
1024 1.79 2.07 1.91 1.72 1.44 1.19 1.05 0.99 –
512 1.83 2.17 2.06 1.92 1.66 1.33 1.13 1.02 0.96
128 1.85 2.23 2.18 2.17 2.01 1.61 1.31 1.12 1.00
1 2 4 8 16 32 64 128 256
batch size B
prompt length L
HF INT8 (bnb LLM.int8)
2048 0.15 0.16 0.17 0.20 0.25 0.36 0.47
1024 0.15 0.15 0.15 0.17 0.20 0.27 0.36
512 0.15 0.15 0.14 0.15 0.16 0.22 0.27
128 0.15 0.14 0.13 0.14 0.14 0.18 0.19
1 2 4 8 16 32 64
batch size B
HF NF4 (bnb)
2048 0.72 0.80 0.78 0.85 0.85 0.97 0.96
1024 0.71 0.77 0.75 0.78 0.76 0.90 0.96
512 0.72 0.77 0.73 0.76 0.70 0.83 0.90
128 0.71 0.75 0.72 0.74 0.66 0.77 0.82
1 2 4 8 16 32 64
batch size B
prompt length L
4×
2×
1×
0.5×
0.25×
end-to-end speedup vs BF16 (same backend)
< 1 slower (red) · > 1 faster (blue)
How to read: each cell is the end-to-end latency of a batch of B requests (prompt L, 128 generated tokens) on BF16 divided by the same on the quantized model, for the same backend. Blue: quantization is faster; red: it is slower; –: out of memory or not measured. GPTQ-INT4 behaves like AWQ-INT4 (same Marlin kernel).